Christian Dane Beels

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Education

United States Military Academy — West Point, NY
B.S. in Mathematical Sciences — Expected 2028
GPA: 4.007

Research interests: Cybersecurity; artificial intelligence; computational mathematics.

Research Experience

Topological Data Analysis for Data Poisoning Prevention

Cadet Researcher — *Jan. 2026 – Present*

Department of Mathematical Sciences, USMA | Advisor: Dr. Joseph Dorta

- Reconstruct and modify a topological data analysis algorithm to investigate detection and removal of poisoned records from network intrusion detection system training data.
- Paper accepted for presentation at the 2026 MIT Undergraduate Research Technology Conference (URTC); unable to attend.
- Received the Outstanding Poster Award at MAA MathFest 2026.

Hunting Hackers with Data

Cadet Researcher — *Aug. 2026 – Present*

Department of Mathematical Sciences, USMA | Advisor: CPT Nolan Hedglin

- Explore data analysis methods to identify adversarial activity on computer networks using data supplied by the Cyber Protection Brigade.

Speech Recognition as a Threat Vector in Real-Time Doxxing

Cadet Researcher — *Sept. – Dec. 2025*

Army Cyber Institute, West Point, NY | Advisor: MAJ Nicholas Harrell, PhD

- Studied the feasibility of adversaries using consumer wearable technology to expose information on U.S. forces in real time.
- Authored a paper on methods, findings, and operational security implications; presented findings to the USMA Department of Electrical Engineering and Computer Science.

Radiation Environment Modeling for Interceptor Survivability

Research Project Intern — *June – July 2025*

Missile Defense Agency, Redstone Arsenal, AL | Advisor: Michael McKenna

- Developed and analyzed nuclear radiation environment models to inform survivability and lethality considerations for anti-ICBM interceptor launch scenarios.
- Presented models and results to a panel of agency and military leadership.

SPICE-Based Modeling of Space Radiation Effects on Satellite Amplifiers

Cadet Researcher — *Jan. – Apr. 2025*

Department of Physics and Nuclear Engineering, USMA | Advisor: LTC Jill Rahon, PhD

- Modeled space radiation effects on RF amplifiers using SPICE simulations to evaluate performance degradation and identify conditions associated with loss of signal integrity.

Presentations

Beels, C. D. (2026). *Topological Data Analysis for Data Poisoning Prevention in Network Intrusion Detection Systems.* Poster presentation, MAA MathFest. Outstanding Poster Award.

Beels, C. D. (May 2026). *Topological Data Analysis for Data Poisoning Prevention in Network Intrusion Detection Systems.* Poster presentation, USMA Math Research Symposium (MaRS).

Beels, C. D. (Dec. 2025). *Speech Recognition as a Threat Vector in Real-Time Doxxing.* Poster presentation, USMA Department of Electrical Engineering and Computer Science.

Beels, C. D., & Ruffing, C. (Apr. 2025). *SPICE-Based Modeling of Space Radiation Effects on Satellite Amplifiers.* Poster presentation, USMA Department of Physics and Nuclear Engineering.

Honors and Awards

Outstanding Poster Award, MAA MathFest	2026
Distinguished Cadet Award, USMA	2026
Master of the Sword Award, USMA	2026
Superintendent's Award, USMA	2025
Dean's List, USMA (four recognitions)	2025, 2026
Norwegian Foot March Badge (Silver)	2025, 2026

Technical Skills

Programming	Python, C, JavaScript, HTML, CSS
Scientific computing	giotto-tda, PyTorch, NumPy, Matplotlib, Wolfram Mathematica
Modeling	SPICE-based circuit simulation

Leadership and Military Experience

Assistant Cadet-in-Charge, West Point Cyber Team Aug. 2025 – Present
Department of Electrical Engineering and Computer Science, USMA

- Lead team participation in collegiate, national, and international cybersecurity competitions; ranked in the top 1% individually and top 0.5% with the team in National Cyber League; qualified for President's Cup Finals, Individual Offensive Track.

Headquarters Platoon Sergeant, Company G-1 Aug. – Dec. 2026

- Responsible for the success and accountability of more than 20 subordinate staff officers.

Cadet Basic Training Squad Leader Aug. – Dec. 2026

- Led and mentored nine new cadets; taught military tactics, rifle marksmanship, land navigation, and character development.

Team Leader, USMA Aug. 2025 – May 2026

- Mentored an underclass cadet in academic, military, physical, and character development.

Air Assault School, 101st Airborne Division June 2025
Fort Campbell, KY; trained in helicopter operations, sling-load operations, and rappelling.

Teaching, Service, and Outreach

Tutor, West Point Center for Enhanced Performance 2024 – Present
Lead evening study sessions before major graded events in single- and multivariable calculus, computer science, chemistry, psychology, and economics.

Volunteer Web Developer, ProVetUSA 2022 – 2023
Built online course materials to support veterans transitioning to civilian employment.

Volunteer, Fuller Center for Housing of Greater NYC 2020 – 2024
Contributed to construction of low-income housing in Westchester, NY.